Example 1.16

Find the rank of the following matrices which are in row-echelon form :

(i) $\begin{bmatrix} 2 & 0 & -7 \\ 0 & 3 & 1 \\ 0 & 0 & 1 \end{bmatrix}$ (ii) $\begin{bmatrix} -2 & 2 & -1 \\ 0 & 5 & 1 \\ 0 & 0 & 0 \end{bmatrix}$ (iii) $\begin{bmatrix} 6 & 0 & -9 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

Solution

(i) Let $A = \begin{bmatrix} 2 & 0 & -7 \\ 0 & 3 & 1 \\ 0 & 0 & 1 \end{bmatrix}$. Then A is a matrix of order 3×3 and $\rho(A) \le 3$

The third order minor $|A| = \begin{vmatrix} 2 & 0 & -7 \\ 0 & 3 & 1 \\ 0 & 0 & 1 \end{vmatrix} = (2)(3)(1) = 6 \neq 0$. So, $\rho(A) = 3$.

Note that there are three non-zero rows.

(ii) Let $A = \begin{bmatrix} -2 & 2 & -1 \\ 0 & 5 & 1 \\ 0 & 0 & 0 \end{bmatrix}$. Then A is a matrix of order 3×3 and $\rho(A) \le 3$.

The only third order minor is $|A| = \begin{vmatrix} -2 & 2 & -1 \\ 0 & 5 & 1 \\ 0 & 0 & 0 \end{vmatrix} = (-2)(5)(0) = 0$. So $\rho(A) \le 2$.

There are several second order minors. We find that there is a second order minor, for example, $\begin{vmatrix} -2 & 2 \\ 0 & 5 \end{vmatrix} = (-2)(5) = -10 \neq 0$. So, $\rho(A) = 2$.

Note that there are two non-zero rows. The third row is a zero row.

(iii) Let $A = \begin{bmatrix} 6 & 0 & -9 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$. Then A is a matrix of order 4×3 and $\rho(A) \le 3$.

The last two rows are zero rows. There are several second order minors. We find that there is a second order minor, for example, $\begin{vmatrix} 6 & 0 \\ 0 & 2 \end{vmatrix} = (6)(2) = 12 \neq 0$. So, $\rho(A) = 2$.

Note that there are two non-zero rows. The third and fourth rows are zero rows.

We observe from the above example that the rank of a matrix in row echelon form is equal to the number of non-zero rows in it. We state this observation as a theorem without proof. ■

Theorem 1.11

The rank of a matrix in row echelon form is the number of non-zero rows in it.

The rank of a matrix which is not in a row-echelon form, can be found by applying the following result which is stated without proof.

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